Staging schema

GravityDW.stg | 12 physical tables

stg.Country

```
PK,I StagingRowID BIGINT NOT NULL
country_id INT NULL
country_name VARCHAR(200) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.Address

```
PK,I StagingRowID BIGINT NOT NULL
address_id INT NULL
country_id INT NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.AddressStatus

```
PK,I StagingRowID BIGINT NOT NULL
status_id INT NULL
address_status VARCHAR(30) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.Customer

```
PK,I StagingRowID BIGINT NOT NULL
customer_id INT NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.BookLanguage

```
PK,I StagingRowID BIGINT NOT NULL
language_id INT NULL
language_code VARCHAR(8) NULL
language_name VARCHAR(50) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.Book

```
PK,I StagingRowID BIGINT NOT NULL
book_id INT NULL
title VARCHAR(400) NULL
isbn13 VARCHAR(13) NULL
language_id INT NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.CustomerAddress

```
PK,I StagingRowID BIGINT NOT NULL
customer_id INT NULL
address_id INT NULL
status_id INT NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.CustomerSurvey

```
PK,I StagingRowID BIGINT NOT NULL
CustomerIDRaw NVARCHAR(100) NULL
DiscoverySourceRaw NVARCHAR(255) NULL
PurchasePreferenceRaw NVARCHAR(255) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.OrderStatus

```
PK,I StagingRowID BIGINT NOT NULL
status_id INT NULL
status_value VARCHAR(20) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.OrderHistory

```
PK,I StagingRowID BIGINT NOT NULL
history_id INT NULL
order_id INT NULL
status_id INT NULL
status_date DATETIME NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.CustOrder

```
PK,I StagingRowID BIGINT NOT NULL
order_id INT NULL
order_date DATETIME NULL
customer_id INT NULL
dest_address_id INT NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

stg.OrderLine

```
PK,I StagingRowID BIGINT NOT NULL
line_id INT NULL
order_id INT NULL
book_id INT NULL
price DECIMAL(5,2) NULL
LoadBatchID INT NOT NULL
ExtractedAt DATETIME2(3) NOT NULL
```

PK = primary key; I = identity. Every table has its own StagingRowID identity primary key.
Source fields remain nullable. Source identifiers are ordinary columns; ETL checks their uniqueness and relationships.
LoadBatchID is explicitly supplied for the shared extraction batch. ExtractedAt defaults to SYSUTCDATETIME().
ExtractedAt records staging insertion time, not order time or survey response time. Full refresh clears staging only.

Target warehouse star schema

GravityDW.dw | 5 dimensions + 1 fact | 40 implemented columns

Grain: one FactOrderLine row per source order_line.line_id. A line is not a book-copy quantity.

dw.DimDate

PK DateKey INT NOT NULL
U* FullDate DATE NULL
CalendarYear SMALLINT NULL
QuarterNumber TINYINT NULL
MonthNumber TINYINT NULL
MonthName NVARCHAR(10) NOT NULL
YearMonth NVARCHAR(7) NOT NULL

dw.DimBook

PK,I BookKey INT NOT NULL
U* SourceBookID INT NULL
Title NVARCHAR(400) NOT NULL
ISBN13 NVARCHAR(13) NULL
LanguageCode NVARCHAR(8) NULL
LanguageName NVARCHAR(50) NOT NULL
LanguageGroup NVARCHAR(30) NOT NULL

dw.FactOrderLine

PK,I OrderLineKey INT NOT NULL
U SourceOrderLineID INT NOT NULL
SourceOrderID INT NOT NULL
FK OrderDateKey INT NOT NULL
FK BookKey INT NOT NULL
FK CustomerKey INT NOT NULL
FK DeliveryCountryKey INT NOT NULL
FK OrderStatusKey INT NOT NULL
OrderLineCount INT NOT NULL
RecordedLinePrice DECIMAL(5,2) NULL
LoadBatchID INT NOT NULL

dw.DimCustomer

PK,I CustomerKey INT NOT NULL
U* SourceCustomerID INT NULL
SurveyResponseAvailable BIT NULL
PurchasePreference NVARCHAR(255) NOT NULL
DiscoverySource NVARCHAR(255) NOT NULL
ActiveCountryCount INT NULL
AddressCountryCategory NVARCHAR(40) NOT NULL
SingleActiveCountryID INT NULL
SingleActiveCountryName NVARCHAR(200) NULL

dw.DimOrderStatus

PK,I OrderStatusKey INT NOT NULL
U* SourceStatusID INT NULL
StatusName NVARCHAR(50) NOT NULL

dw.DimDeliveryCountry

PK,I DeliveryCountryKey INT NOT NULL
U* SourceCountryID INT NULL
CountryName NVARCHAR(200) NOT NULL

PK = primary key; FK = enforced foreign key; I = identity; U = unique; U* = unique when the value is not NULL.

Double bars indicate exactly one; a circle plus a crow’s foot indicates zero or many.
Each order line references exactly one member of each dimension. Each dimension member may be referenced by zero or many order lines.
Key 0 is reserved for unknown dimension members. SourceOrderID is a non-unique order identifier, not a foreign key.

Customer/survey/address attributes are current Type 1 values. SingleActiveCountryID is a source ID, not a warehouse FK.